

Steve Cabrel Tegua Kouam
Department of Physics, Faculty of Science
University of Douala, Cameroon
steve.tegua@facsciences-uy1.cm

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Editor-in-Chief
Quantum Science and Technology
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Bristol BS1 6HG, UK

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “**Quantum-Enhanced Agrivoltaic Digital Twin: Non-Markovian Coherence Control, NPoM Plasmonic Diagnostics, and BB84-Secured IoT Telemetry**” for consideration as a Research Article in *Quantum Science and Technology*.

Publication route

We explicitly confirm that we opt to publish this manuscript via the **standard subscription route (carrying zero Author Processing Charge, \$0 APC)**.

Summary of the work

Coherence in photosynthetic light-harvesting complexes is well documented, yet no operating device has used it as a design variable. We present the first quantum digital twin for agrivoltaics that does so: a 9-site Fenna–Matthews–Olson (FMO) complex is coupled to a nanoparticle-on-mirror (NPoM) plasmonic nanocavity inside an organic photovoltaic (OPV) stack, forming a 9×9 dressed Hamiltonian propagated with numerically exact non-Markovian process tensor dynamics (PT-HOPS/SBD). Dual-band OPV spectral filtering extends FMO coherence by 50% at room temperature; single-molecule SERS diagnostics are retained; microclimate (FAO-56), lifecycle assessment, and BB84-secured IoT telemetry close the system. The contribution to *QST* is method-led: numerically exact open-quantum-system dynamics used as a design tool for a plasmonic–biological device, with the strong-coupling regime confined to a 1% sentinel fraction so that single-molecule quantum sensing coexists with near-baseline transport (97.1% global canopy yield).

Our main scientific contributions are:

- **Numerically exact non-Markovian dynamics as a design tool:** we propagate the 9-site dressed Hamiltonian with PT-HOPS/SBD ($L = 8$, $K = 2$, $n = 100$) through a validated convergence hierarchy. The results locate an optimal NPoM mode volume ($V_{\text{mode}} = 1.2 \text{ nm}^3$): dual-band filtering extends FMO coherence by 50% and transport yields by 25% at room temperature, while the collective strong coupling ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$) stays confined to a 1% sentinel fraction that preserves 97.1% of the global canopy yield.
- **Single-molecule quantum sensing coexisting with transport:** the same strong-coupling regime that quenches local transport provides Purcell-enhanced single-molecule SERS readout (EF up to 1.6×10^3 across the scan), quantifying the plasmonic trade-off that quantum-engineered photonic environments must respect.
- **System closure with quantum-secured telemetry:** FAO-56 microclimate modeling, lifecycle assessment, and room-temperature BB84 QKD ($N_{\text{raw}} \approx 1200$ bits, 3.7 ms edge inference on ARM Cortex-M4) integrate the quantum core into a field-scale, cybersecured deployment scenario.

Why *Quantum Science and Technology*?

Quantum Science and Technology publishes cross-disciplinary advances spanning quantum photonics, quantum biology, quantum sensing, and quantum communication. This manuscript sits at their intersection: exact open-quantum-system dynamics (quantum biology) drives the design of a plasmonic nanocavity device (quantum photonics) whose strong-coupling regime doubles as a single-molecule sensor (quantum sensing), integrated with quantum-keyed telemetry (quantum communication) and validated against field-scale constraints. Few submissions traverse all four; the physics content — engineered photonic environments, strong-coupling trade-offs, process tensor methods at production scale — is the core of the paper, with the agricultural application as its proving ground.

Prior discussions and related submissions

A companion manuscript addressing selective vibronic excitation in isolated FMO complexes is under revision at the *Journal of Physical Chemistry Letters* (Manuscript ID: jz-2026-00994t). The present manuscript extends that quantum framework to a complete agrivoltaic digital twin incorporating NPoM cavity electrodynamics, SERS diagnostics, microclimate physics, lifecycle assessment, and BB84 cybersecurity. No part of this manuscript is under consideration elsewhere.

All authors have approved the manuscript and agree with its submission to *Quantum Science and Technology*. The authors declare no competing financial or non-financial interests.

Suggested reviewers:

1. **Prof. Shaul Mukamel** — University of California, Irvine (smukamel@uci.edu) — *Non-linear optical spectroscopy, open quantum systems.*
2. **Prof. Jeremy J. Baumberg** — University of Cambridge (jjb12@cam.ac.uk) — *NPoM nanoplasmonics, sub-nanometer strong coupling, SERS.*
3. **Prof. Alán Aspuru-Guzik** — University of Toronto (aspuru@utoronto.ca) — *Quantum biology, environment-assisted transport, digital twins.*
4. **Prof. Alexandra Olaya-Castro** — University College London (a.olaya@ucl.ac.uk) — *Biomolecular quantum dynamics, vibronic coherence.*
5. **Prof. Greg D. Scholes** — Princeton University (gscholes@princeton.edu) — *Quantum coherence in light-harvesting architectures.*

Journal cover art proposal

Upon acceptance of this manuscript, we would be delighted to submit a high-resolution 3D artistic rendering illustrating the multi-scale quantum agrivoltaic digital twin—depicting sub-nanometer NPoM plasmonic cavities coupled to crop light-harvesting complexes under a spectrally engineered OPV canopy—for consideration as a front cover image for *Quantum Science and Technology*.

Sincerely,

Steve Cabrel Tegua Kouam
(On behalf of all authors)

Enclosures: Manuscript (Manuscript_QST_26-07-29.pdf), Supporting Information (SI.pdf), and Figures.